```
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Import File From Computer

```
from google.colab import files
files.upload()
```

```
df_housing = pd.read_csv('housing.csv')
```

```
!unzip data.zip  #when the file is zipped
```

Import File From Drive

```
from google.colab import drive
drive.mount('/content/gdrive')
```

```
Mounted at /content/gdrive
```

```
path='/content/gdrive/MyDrive/AI_ML_Research_Papers_2023_2026.csv'
df_Research_Paper = pd.read_csv(path)
```

Access Data

https://www.kaggle.com/

https://archive.ics.uci.edu/dataset/240/human+activity+recognition+using+smartphones

```
from sklearn import datasets

dir(datasets)
```

Dataset Analysis

```
df_Research_Paper.columns
```

```
df_Research_Paper.head()
```

```
df_Research_Paper.info()
```

```
df_housing.describe()
```

Working with Missing Value

```
#Check if there is any missing value
df_Research_Paper.isnull().values.any()
```

```
# Check which columns have the missing value
df_Research_Paper.isnull().sum()
```

```
# Check the rows with missing values
df_Research_Paper[df_Research_Paper.isnull().any(axis=1)]
```

```
# Removing the rows with missing value
df_Research_Paper = df_Research_Paper.dropna()
```

```
# Check if there is missing values anymore
df_Research_Paper.isnull().values.any()
```

```
df_housing.head()
```

```
df_housing.info()
```

```
df_housing['total_bedrooms'].fillna(df_housing['total_bedrooms'].mean(), inplace=True)
```

```
df_housing.info()
```

Data Slicing

```
df_housing.describe()
```

```
df_housing.iloc[9:25, 2:5]
```

```
df_housing.loc[9:25, 'housing_median_age':'total_rooms']
```

Data Labeling

```
labels = df_Research_Paper['topic']
labels.unique()
```

```
label_map = {
    'Machine Learning': 0,
    'LLMs': 1,
    'RAG': 2,
    'AI Agents': 3,
    'Diffusion Models': 4
}

df_Research_Paper['label'] = df_Research_Paper['topic'].map(label_map)
```

```
df_Research_Paper[['topic','label']].tail()
```

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()
df_Research_Paper['label'] = le.fit_transform(df_Research_Paper['topic'])

mapping = dict(zip(le.classes_, range(len(le.classes_))))
print(mapping)
```

Feature & Target

```
df_housing.head()
```

```
features = df_housing.drop('median_house_value', axis=1)
target = df_housing['median_house_value']
```

```
X = df_housing[features.columns].copy()
y = df_housing['median_house_value'].values.reshape(-1, 1)
```

```
X.shape, y.shape
```

Box Plot for Visualization

```
plt.figure(figsize=(6,4))
sns.boxplot(target)
plt.title("Box Plot of Feature")
plt.show()
```